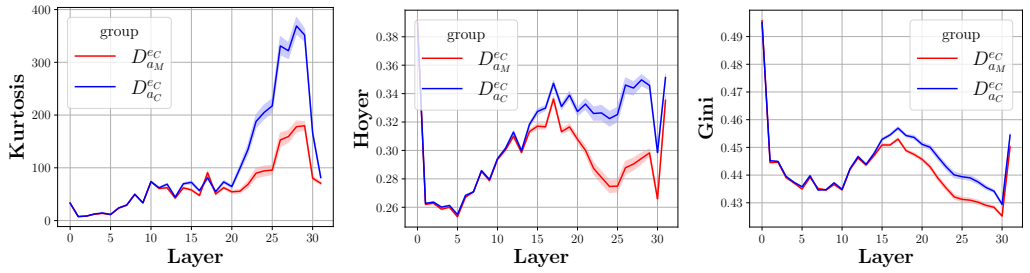

Figure 2: Skewness of the hidden state activations of Llama3-8B when in presence of knowledge conflicts. Blue and red lines represent the skewness of hidden states from $D^{ec}_{a_C}$ and $D^{ec}_{a_M}$, respectively. Higher scores indicate a more skewed distribution. Additional analyses are available in Appendix C.

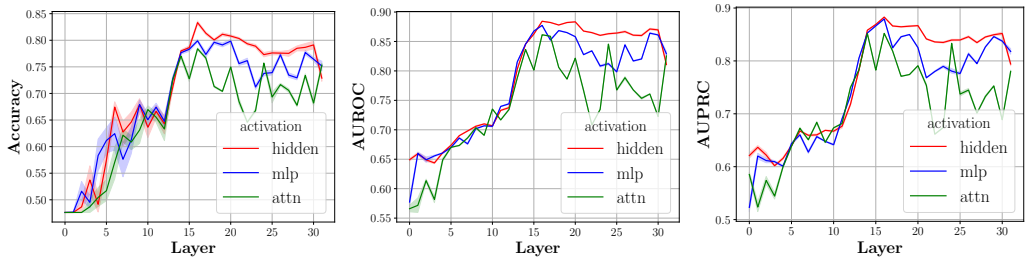

Figure 3: Accuracy, AUROC, and AUPRC of probing models on predicting which source of knowledge the model will use to predict the answer in Llama3-8B. More results are Skewness of the hidden state activations of Llama3-8B when the model uses knowledge from different sources to predict the answer. Additional results are available in Appendix E.

align with a_M. The model uses contextual knowledge and parametric knowledge to answer the questions from $D^{ec}_{a_C}$ and $D^{ec}_{a_M}$, respectively.

First, we examine the residual streams' distribution patterns in the two groups of instances $D^{ec}_{a_C}$ and $D^{ec}_{a_M}$. We measure the skewness of the residual stream using Kurtosis, Hoyer and Gini index. We present the results of NQSwap using Llama3-8B in Figure 2, and more results are provided in the Appendix C. We find that when the model uses contextual knowledge for prediction ($D^{ec}_{a_C}$, blue lines shown in Figure 2), the residual stream shows a significantly skewed distribution compared with using parametric knowledge from the 20th to 30th layers. Therefore, the distribution patterns of the residual stream can indicate the model will use different sources of knowledge. It provides the foundation for predicting the model's behaviour in advance, which can be used to mitigate the generation of undesirable responses in advance.

Based on the above observation, we probe the residual stream to analyse the possibility of predicting which source of knowledge will be used to generate the answer. The probing model is trained to classify whether the model will generate a_C or a_M based on the activation from $D^{ec}_{a_C}$ or $D^{ec}_{a_M}$. We present the probing results in Figure 3. We observe that the probing model's performance gradually improves from the first layer to the 16th layer, which occurs after the signal of knowledge conflict has already reached its peak at the 13th and 14th layers. This observation suggests that the decision of which knowledge to use occurs after the detection of the knowledge conflict signal.

5 Related Work

Contextual and parametric knowledge conflict can happen when the retrieved external knowledge in the context does not agree with the parametric knowledge which is memorised during pre-training [Longpre et al., 2021, Xu et al., 2024, Xie et al., 2024, Su et al., 2024, Wang et al., 2023, Mallen et al., 2023]. Previous works found models may prefer the contextual knowledge [Wang et al., 2023, Su et al., 2024, Xie et al., 2024, Ortu et al., 2024] when parametric and contextual knowledge conflicts, and the relevance, length, and the number of the evidence will influence the model's preferences [Xie et al., 2024, Su et al., 2024]. To detect the conflict, previous work [Wang et al.,

2023] designed a multi-step prompting strategy to detect the knowledge, which involves parametric knowledge generation, fine-grained sentence consistency checking, and potential conflict reduction. However, this pipeline significantly reduces efficiency and lacks an understanding of the mechanism of how LLMs detect and resolve conflict.

6 Conclusions

In this work, we analyse the residual stream of the language models when context-parameter knowledge conflicts. First, we find that LLMs exhibit internal mechanisms for identifying conflicts in the mid-layers. Second, we find that the residual stream shows distinct skewness patterns when the model uses context and parametric knowledge to predict. Our analysis provides insights into the behaviour of LLMs in the presence of knowledge conflicts. This work is the preliminary study of "Steering Knowledge Selection Behaviours in LLMs via SAE-Based Representation Engineering [Zhao et al., 2024b]", where we propose a training-free inference-time intervention method SPARE to steer the knowledge selection behaviours of LLMs under context-memory knowledge conflict.

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References

Zeyuan Allen-Zhu and Yuanzhi Li. Physics of language models: Part 1, context-free grammar. *arXiv preprint arXiv:2305.13673*, 2023.

Tom B Brown. Language models are few-shot learners. *arXiv preprint arXiv:2005.14165*, 2020.

Canyu Chen and Kai Shu. Can llm-generated misinformation be detected? *arXiv preprint arXiv:2309.13788*, 2023a.

Canyu Chen and Kai Shu. Combating misinformation in the age of llms: Opportunities and challenges. *AI Magazine*, 2023b.

Alexis Conneau, German Kruszewski, Guillaume Lample, Loïc Barrault, and Marco Baroni. What you can cram into a single vector: Probing sentence embeddings for linguistic properties. *arXiv preprint arXiv:1805.01070*, 2018.

Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Amy Yang, Angela Fan, et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, et al. A mathematical framework for transformer circuits. *Transformer Circuits Thread*, 1(1):12, 2021.

Giwon Hong, Jeonghwan Kim, Junmo Kang, Sung-Hyon Myaeng, and Joyce Jiyoung Whang. Why so gullible? enhancing the robustness of retrieval-augmented models against counterfactual noise. In Kevin Duh, Helena Gómez-Adorno, and Steven Bethard, editors, *Findings of the Association for Computational Linguistics: NAACL 2024, Mexico City, Mexico, June 16-21, 2024*, pages 2474–2495. Association for Computational Linguistics, 2024. doi: 10.18653/V1/2024.FINDINGS-NAACL.159. URL https://doi.org/10.18653/v1/2024.findings-naacl.159.

Albert Q Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, et al. Mistral 7b. *arXiv preprint arXiv:2310.06825*, 2023.

Vladimir Karpukhin, Barlas Oguz, Sewon Min, Patrick S. H. Lewis, Ledell Wu, Sergey Edunov, Danqi Chen, and Wen-tau Yih. Dense passage retrieval for open-domain question answering. In Bonnie Webber, Trevor Cohn, Yulan He, and Yang Liu, editors, *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing, EMNLP 2020, Online, November 16-20, 2020*, pages 6769–6781. Association for Computational Linguistics, 2020. doi: 10.18653/V1/2020.EMNLP-MAIN.550. URL https://doi.org/10.18653/v1/2020.emnlp-main.550.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, et al. Retrieval-augmented generation for knowledge-intensive nlp tasks. *Advances in Neural Information Processing Systems*, 33: 9459–9474, 2020.

Shayne Longpre, Kartik Perisetla, Anthony Chen, Nikhil Ramesh, Chris DuBois, and Sameer Singh. Entity-based knowledge conflicts in question answering. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 7052–7063, Online and Punta Cana, Dominican Republic, November 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.565. URL https://aclanthology.org/2021.emnlp-main.565.

Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khashabi, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In Anna Rogers, Jordan L. Boyd-Graber, and Naoaki Okazaki, editors, *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), ACL 2023, Toronto, Canada, July 9-14, 2023*, pages 9802–9822. Association for Computational Linguistics, 2023. doi: 10.18653/V1/2023.ACL-LONG.546. URL https://doi.org/10.18653/v1/2023.acl-long.546.

Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, et al. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022.

Francesco Ortu, Zhijing Jin, Diego Doimo, Mrinmaya Sachan, Alberto Cazzaniga, and Bernhard Schölkopf. Competition of mechanisms: Tracing how language models handle facts and counterfactuals. *arXiv preprint arXiv:2402.11655*, 2024.

Fabio Petroni, Tim Rocktäschel, Sebastian Riedel, Patrick S. H. Lewis, Anton Bakhtin, Yuxiang Wu, and Alexander H. Miller. Language models as knowledge bases? In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing, EMNLP-IJCNLP 2019, Hong Kong, China, November 3-7, 2019*, pages 2463–2473. Association for Computational Linguistics, 2019. doi: 10.18653/V1/D19-1250. URL https://doi.org/10.18653/v1/D19-1250.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. *Advances in Neural Information Processing Systems*, 36, 2024.

Zhaochen Su, Jun Zhang, Xiaoye Qu, Tong Zhu, Yanshu Li, Jiashuo Sun, Juntao Li, Min Zhang, and Yu Cheng. Conflictbank: A benchmark for evaluating the influence of knowledge conflicts in llm. *arXiv preprint arXiv:2408.12076*, 2024.

Gemini Team, Rohan Anil, Sebastian Borgeaud, Yonghui Wu, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, et al. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*, 2023.

Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023.